

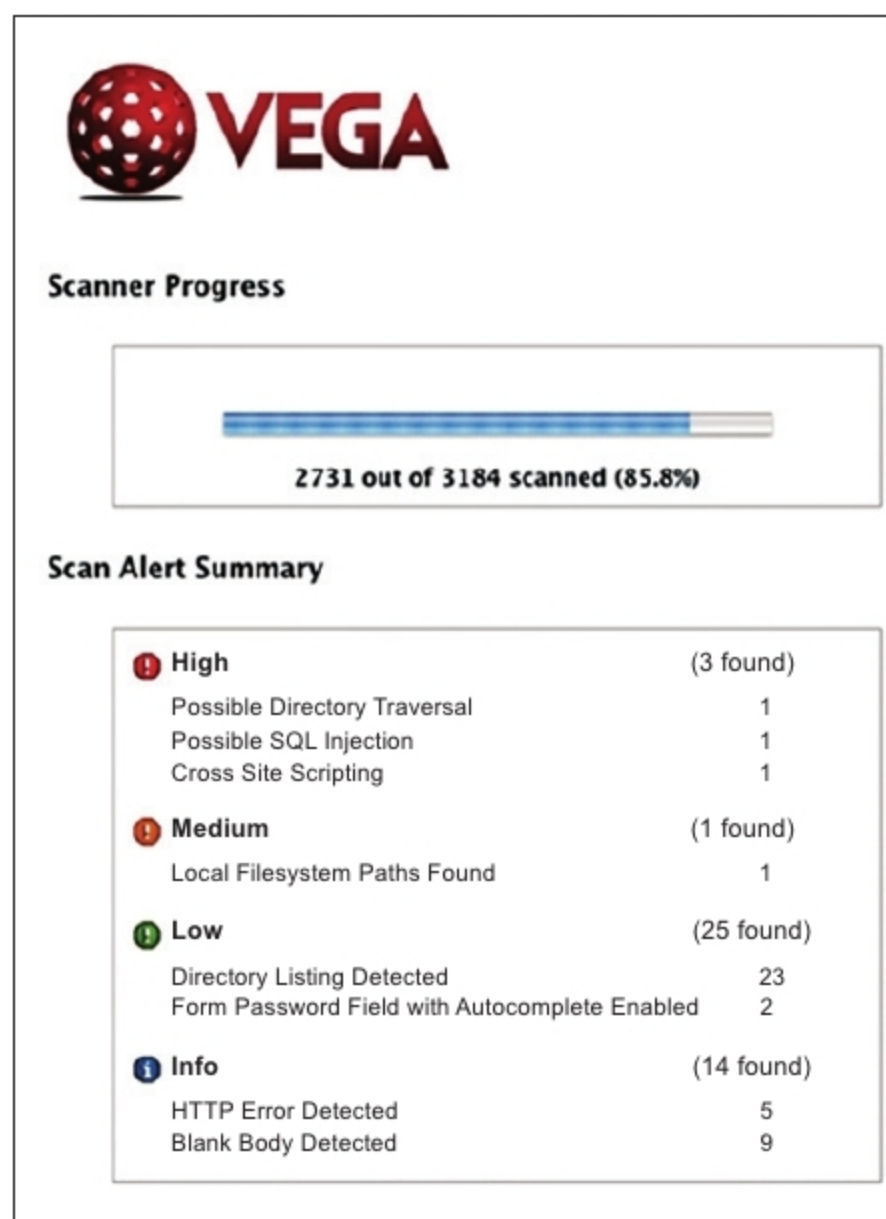


Figure 3: Deep analysis in the Vega scanner

Table 2: Penetration testing and audit tools

Tool	URL
AAPT	https://androidaapt.com/
OWASP Zed Attack Proxy	https://www.owasp.org
QARK (Quick Android Review Kit)	https://github.com/linkedin/qark
Devknox	https://devknox.io/
Drozer	https://labs.mwrinfosecurity.com/tools/drozer
MobSF (Mobile Security Framework)	https://github.com/MobSF/Mobile-Security-Framework-MobSF
Mitmproxy	https://mitmproxy.org/
iMAS	https://github.com/project-imas/about

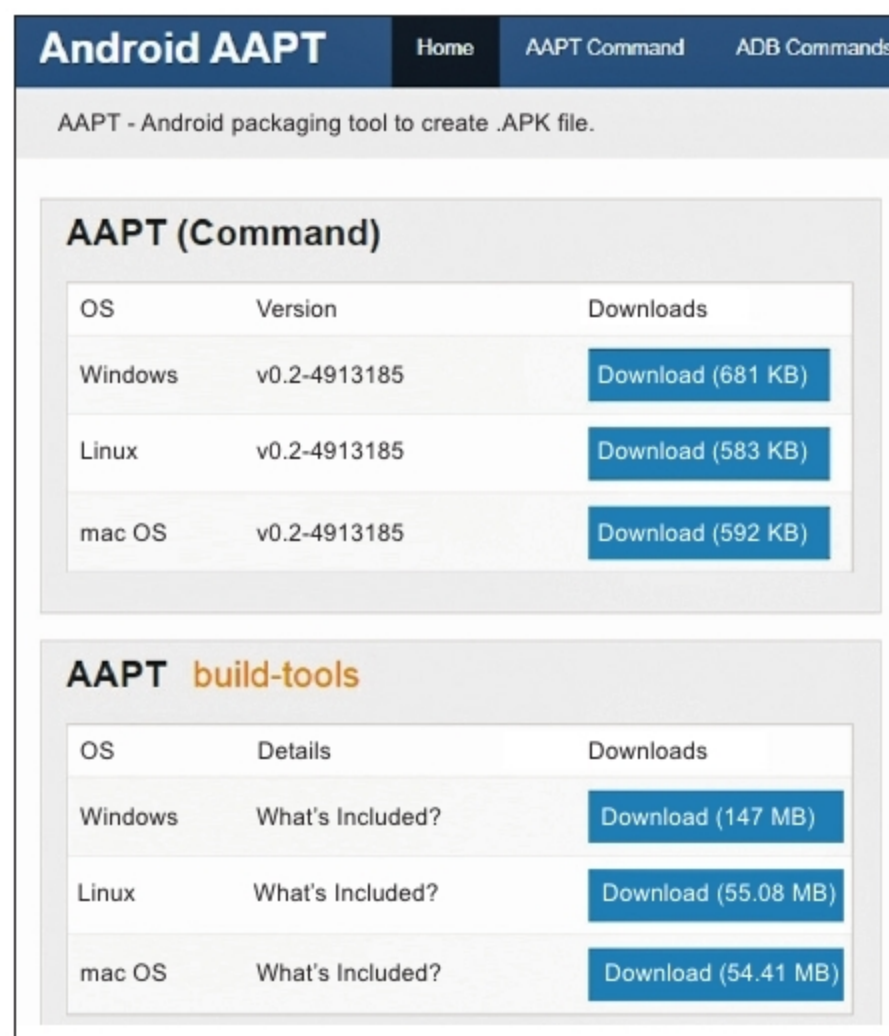


Figure 4: Official portal of Android Asset Packaging Tool (AAPT)



Figure 5: Fetching APK permissions using AAPT

but directly download the APK from different portals. This practice can be quite dangerous; so users are advised to use advanced tools and analyse the inherent permissions before installing the app on the mobile phone.

In addition, research scholars and practitioners can work on this domain of smartphone malware analysis by collecting the permissions from assorted APK files, and then training the data sets using machine learning. This will help to predict the security, privacy and data breach issues of upcoming smartphone apps. **END** 🐧

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